

Summary of Housing Price Prediction Project

The dataset consists of features such as area, bedrooms, bathrooms, parking, prefarea, hot water heating, air conditioning, furnishing status, guestroom and main road. The dataset also includes the corresponding price of the houses to train the model before making any predictions. The dataset is small, it includes 545 rows and 13 columns.

In the data preprocessing and cleaning phase, no missing values were found in the dataset. The categorical variables (Yes/No) were converted into numerical variables (1/0) using One Hot Encoding as Machine Learning concepts can only be applied to numerical data. During the data preprocessing phase, all the columns except Price were taken as feature variable because the relationships between target variable and independent variable were unknown, so no features were removed based on correlation.

The encoded data is split into training and testing sets of 80:20 ratio respectively. A Linear Regression model was trained to predict house prices, the model achieved an MAE of 9.70043×10^5 , RMSE of 1.324×10^6 and R^2 score of 0.652924. This indicates the model's ability to explain 65% of variation in house prices and provides reasonably accurate prediction. When a Random Forest model is trained on the same dataset, the R^2 score is 0.612865, MAE is 1.016168×10^6 and RMSE is 1.398857×10^6 .

Linear Regression outperformed Random Forest on this dataset, achieving a higher R^2 and lower MAE and RMSE. This may be due to the small dataset size (545 rows), where Random Forest with default hyperparameters did not generalize as well. Tuning Random Forest hyperparameters did not improve performance, therefore, Linear Regression is better suited for this dataset size.

Price is most influenced by Area(0.54 correlation), Air Conditioning(0.45 correlation), Bathrooms(0.52 correlation) and Stories(0.42 correlation). Unfurnished houses have a negative correlation with Price. Furnished houses are comparatively more expensive than semi-furnished and unfurnished houses.

Based on the visualisations, a real estate business should focus on marketing larger homes with premium amenities like air conditioning and better furnishing.